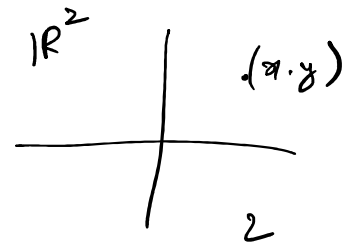
LECTURE 20

07 November 2023 11:41

Functions of several variables: Two variables:

$f(x,y) = x^2+y^2$ on

Domain of definition: whole of $\mathbb{R}^2$



$f(x,y) = \dfrac{xy}{x^2+y^2}$ $\quad \mathbb{R}^2 \setminus \{0,0\}$ $\quad \mathbb{C} = \mathbb{R}^2$

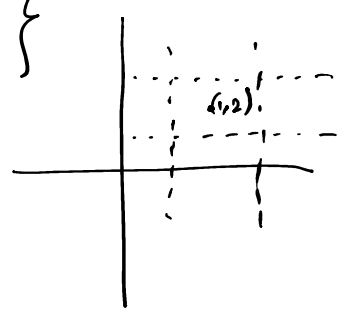
$f(x,y) = \dfrac{xy}{x^2-y^2}$ $\quad \mathbb{R}^2 \setminus \{y = \pm x\}$

$A = \{(x,y) : \underbrace{(x-1)^2+(y-2)^2 < 9}\}$ → open ball in $\mathbb{R}^2$

→ circular nbd of $(1,2)$

$B = \{(x,y) : |x-1| < \frac{1}{2}$ & $|y-2| < 1\}$

Rect. nbd of $(1,2)$ $\quad \frac{1}{2} < x < \frac{3}{2}$ $\quad 1 < y < 3$



Limit of a f^n. of several variables:

$$\lim_{(x,y)\to(a,b)} f(x,y) = l$$

Eg. 1) $f(x,y) = xy^2 + yx^2 + x^3 + y^3$

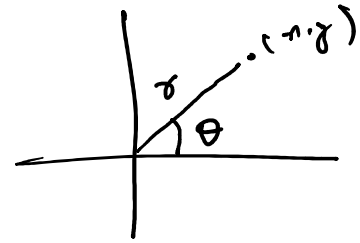
$$\lim_{(x,y)\to(0,0)} f(x,y) = 0$$

2) $$\lim_{(x,y)\to(0,0)} \frac{xy(x^2-y^2)}{x^2+y^2}$$

$x = r\cos\theta \quad y = r\sin\theta$

$(r,\theta) \to (0,0)$

$$= \lim_{(r,\theta)\to(0,0)} r^2 \sin\theta \cos\theta \,(\cos^2\theta - \sin^2\theta)$$

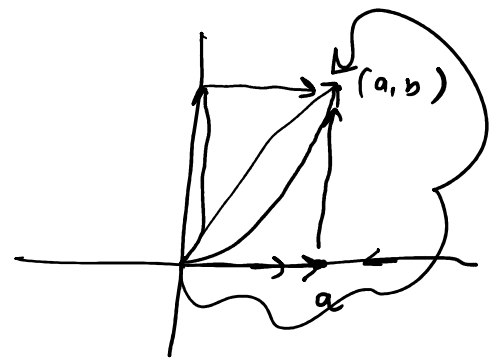


$$= \frac{r^2}{2} \underbrace{2\sin\theta\cos\theta}_{\sin 2\theta} \cos 2\theta$$

$$= \frac{r^2}{4} 2\sin 2\theta \cos 2\theta = \lim_{(r,\theta)\to(0,0)} \frac{r^2}{4} \sin 4\theta = 0$$

<u>Note</u>: If in polar form: r^p $p>0$ remains in numerator & no θ in denominator then limit will exist.

3) $f(x,y) = \dfrac{x^2-y^2}{x^2+y^2}$



Along x axis $y=0$ $\lim_{x\to 0} f(x,0) = 1$

Along y axis $x=0$ $\lim_{y\to 0} (0,y) = -1$

$\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

$y = mx$

$$f(x,mx) = \frac{x^2-m^2x^2}{x^2+m^2x^2} = \frac{1-m^2}{1+m^2}$$

$\lim_{x\to a} f(x,y)$ if $\lim f(x,y)$ depend on m → limit does not exist

4) $f(x,y) = \dfrac{xy^2}{x^2+y^4}$

Along x axis $\lim_{x\to 0} f(x,y) = 0$

along y axis $\lim f(x,y) = 0$

Along y axis $\lim_{y \to 0} f(x,y) = 0$

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4} = \lim_{x\to 0} \frac{x \cdot m^2x^2}{x^2+m^4x^4} = \lim_{x\to 0} \frac{m^2x}{1+m^4x^2} = 0$$

$x = my^2$

$$\lim_{y\to 0} \frac{my^2 \cdot y^2}{m^2y^4+y^4} = \lim_{y\to 0} \frac{m\cancel{y^4}}{(m^2+1)\cancel{y^4}} = \frac{m}{m^2+1}$$ depends on m.

5) $f(x,y) = \dfrac{x^2}{x^2+y^2}$

$$\lim_{x\to 0}\lim_{y\to 0} f(x,y) = \lim_{x\to 0} \frac{x^2}{x^2} = 1$$

$$\lim_{y\to 0}\lim_{x\to 0} f(x,y) = \lim_{y\to 0} 0 = 0$$

$\lim_{x\to a}\lim_{y\to b} f(x,y)$ & $\lim_{y\to b}\lim_{x\to a} f(x,y)$ Repeated or Iterated limits

$$\lim_{(x,y)\to(0,0)} \frac{x^2}{x^2+y^2} = \lim_{x\to 0} \frac{x^2}{x^2+m^2x^2} = \frac{1}{1+m^2}$$

$y = mx$

Note: If repeated limits are not equal, simultaneous limit cannot exist

6) $f(x,y) = \dfrac{xy}{x^2+y^2}$

$y = mx$

$$f(x,y) = \frac{mx^2}{x^2+m^2x^2} = \frac{m}{1+m^2}$$

$$\lim_{x\to 0}\lim_{y\to 0} f(x,y) = \lim_{x\to 0}\lim_{y\to 0} \frac{xy}{x^2+y^2} = 0$$

$$\lim_{y\to 0}\lim_{x\to 0} \frac{xy}{x^2+y^2} = 0$$

7) $$\lim_{(x,y)\to(0,0)} x \sin\frac{1}{y} = 0$$

$$\lim_{y\to 0} x \sin\frac{1}{y} = \text{does not exist.}$$

$$\lim_{x\to 0} x \sin\frac{1}{y} = 0$$

8) $$f(x,y) = \frac{xy^3}{x^2+y^6} \longrightarrow$$

$x = my^3$

$$\frac{my^6}{m^2y^6+y^6} = \frac{m}{m^2+1}$$